

EXTERIOR LIGHTING SYSTEM

High-Level Design Document

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Status: Approved for Engineering Analysis

Classification: Synthetic Demonstration Data

1. System Overview

The Exterior Lighting System coordinates its functions through typed AUTOSAR interfaces and ports. Software components exchange signals through provider and consumer ports.

2. Software Architecture

The primary software components are LightingManager, HeadlampControl, IndicatorControl, LightSensorManager, and SwitchControl.

3. Software Components

Component	Role	Primary Responsibility
LightingManager	Coordinator	Combines switch and sensor inputs into lamp commands.
HeadlampControl	Consumer	Drives low and high beam outputs.
IndicatorControl	Consumer	Drives turn indicators and hazard lights.
LightSensorManager	Provider	Provides ambient light level.
SwitchControl	Provider	Reads the column switches.

4. Interfaces

Interface	Provider	Consumer	Signals
IAmbientLight	LightSensorManager	LightingManager	AmbientLightLevel
ISwitchPosition	SwitchControl	LightingManager	SwitchPosition
IHeadlampCommand	LightingManager	HeadlampControl	HeadlampCommand
IIndicatorCommand	LightingManager	IndicatorControl	IndicatorCommand

5. Ports

Component	Port	Direction	Interface
LightSensorManager	AmbientLight_Out	P-PORT	IAmbientLight
LightingManager	AmbientLight_In	R-PORT	IAmbientLight
SwitchControl	SwitchPosition_Out	P-PORT	ISwitchPosition
LightingManager	SwitchPosition_In	R-PORT	ISwitchPosition
LightingManager	HeadlampCommand_Out	P-PORT	IHeadlampCommand
HeadlampControl	HeadlampCommand_In	R-PORT	IHeadlampCommand
LightingManager	IndicatorCommand_Out	P-PORT	IIndicatorCommand
IndicatorControl	IndicatorCommand_In	R-PORT	IIndicatorCommand

6. Signals

Signal	Data Type	Source	Destination
AmbientLightLevel	uint16	LightSensorManager	LightingManager
SwitchPosition	uint8	SwitchControl	LightingManager
HeadlampCommand	uint8	LightingManager	HeadlampControl
IndicatorCommand	uint8	LightingManager	IndicatorControl

7. Dependencies

Source	Relationship	Target	Reason
LightSensorManager	PROVIDES_INTERFACE	IAmbientLight	Publishes ambient light level.
LightingManager	REQUIRES_INTERFACE	IAmbientLight	Consumes ambient light level.
SwitchControl	PROVIDES_INTERFACE	ISwitchPosition	Publishes switch position.
LightingManager	REQUIRES_INTERFACE	ISwitchPosition	Consumes switch position.
LightingManager	PROVIDES_INTERFACE	IHeadlampCommand	Publishes headlamp command.
HeadlampControl	REQUIRES_INTERFACE	IHeadlampCommand	Consumes headlamp command.
LightingManager	PROVIDES_INTERFACE	IIndicatorCommand	Publishes indicator command.
IndicatorControl	REQUIRES_INTERFACE	IIndicatorCommand	Consumes indicator command.

8. Functional Flows

Flow F-001: Ambient Light Level Distribution

LightSensorManager publishes AmbientLightLevel through IAmbientLight. LightingManager receives the signal through AmbientLight_In and updates its internal state model.

Flow F-002: Switch Position Distribution

SwitchControl publishes SwitchPosition through ISwitchPosition. LightingManager receives the signal through SwitchPosition_In and updates its internal state model.

Flow F-003: Headlamp Command Distribution

LightingManager publishes HeadlampCommand through IHeadlampCommand. HeadlampControl receives the signal through HeadlampCommand_In and updates its internal state model.

Flow F-004: Indicator Command Distribution

LightingManager publishes IndicatorCommand through IIndicatorCommand. IndicatorControl receives the signal through IndicatorCommand_In and updates its internal state model.

9. Integration Constraints

All provider and consumer ports shall reference compatible interfaces. Signal names shall remain consistent between interface definitions and functional flows. Architecture dependencies must be traceable to an approved source section.

10. Engineering Traceability

This synthetic HLD supports extraction, retrieval, validation and revision comparison experiments.